

SUBASH KUMAR

kumarsubash423@gmail.com •  +917564977193 • [LinkedIn](#) • [LeetCode](#) • [GitHub](#)

Summary

Backend & AI-capable Software Engineer and IIT Delhi graduate with Almost 2 years of production experience at Sprinklr India Pvt. Ltd. designing scalable microservices, high-throughput REST APIs, and event-driven systems using Java and Spring Boot. Proficient in distributed caching, query optimization with MongoDB, Elasticsearch, and Redis, as well as Data Structures, Algorithms, and System Design fundamentals.

Education

Indian Institute of Technology Delhi
B.Tech in Electrical Engineering

2018 – 2022
CGPA: 7.52/10

Work Experience

Sprinklr India Pvt. Ltd., Gurgaon
Backend Software Engineer

July 2022 – August 2023

- Designed and maintained high-throughput **Java / Spring Boot** microservices handling social publishing and scheduling workflows across 5+ platforms, serving thousands of concurrent users.
- Engineered and integrated secure **REST APIs** for Facebook, Instagram, LinkedIn, and X, implementing robust webhook ingestion, token lifecycles, and **rate-limiting** mechanisms.
- Optimized distributed data pipelines and caching layers using **MongoDB, Elasticsearch, and Redis**, formulating strategic indexing and query tuning to reduce latency by **30%**.
- Developed a standardized automated testing suite using **JUnit, Mockito, and TDD** principles, raising code coverage above **85%** across production services and adopted by 15+ engineers.
- Implemented asynchronous processing and thread pool optimizations for background task scheduling, significantly decreasing API response times during peak traffic hours.

Promax Legal Solution Pvt. Ltd., Mohali
Software Development Intern

May 2021 – July 2021

- Built an automated NLP semantic search pipeline using **Word2Vec** and **FastText** vector embeddings for patent prior-art classification, utilizing architectures foundational to modern **RAG** systems.
- Implemented a **Gaussian Naive Bayes** classifier using Python and NLP pipelines for high-accuracy.
- Automated core analytical query generation workflows, improving patent analyst turnaround efficiency by **90%**.

Career Break – Technical Upskilling & UPSC Preparation

Sep 2023 – May 2026

- Solved **500+ DSA problems** across LeetCode and competitive programming platforms, focusing on Graph Theory, Dynamic Programming, Red-Black Trees, and Advanced Algorithms.
- Refreshed modern backend engineering practices—including LLD, HLD, and microservice scalability—while exploring **Generative AI** integrations, **LLM APIs**, and basic **RAG** architectures.

Technical Skills

Languages:	Java (8/11/17), Python, TypeScript, JavaScript, SQL, C++
Frontend & Web:	React.js, HTML5/CSS3, Tailwind CSS
Backend & APIs:	Spring Boot, Node.js, Express, RESTful APIs, GraphQL, Microservices
AI & Data:	Generative AI, LLMs (OpenAI), LangChain, RAG, NLP, Vector DBs (Pinecone)
Databases & Cloud:	MongoDB, PostgreSQL, Elasticsearch, Redis, AWS (EC2, S3), Kafka
DevOps & Tools:	Docker, Kubernetes (Basics), CI/CD, Git, JUnit 5, Mockito

Projects

Processing Complex 3-D Shapes

Java, Red-Black Trees, Graph Algorithms

- Modeled 3-D geometric mesh structures utilizing custom Graph representations and triangular boundary units.
- Integrated **Red-Black Trees** for optimal $O(\log N)$ spatial coordinate indexing and lookups.
- Implemented **DSU, BFS, DFS, and Dijkstra's** algorithms for mesh connectivity and shortest-path queries.

Concurrent Buyer-Seller Trading Engine

Java, Concurrency, System Design

- Built a multi-threaded electronic marketplace matching engine simulating asynchronous order executions.
- Implemented synchronized thread-safe structures and custom **Priority Queues** to prevent race conditions.

Student Database Management System

Data Structures and Algorithms

- Engineered a custom Hash Table-backed database equipped with Double Hashing and BST-based Separate Chaining to efficiently resolve data collisions.
- Conducted empirical performance profiling to compare both collision strategies and optimize key-based search latency.

Achievements

- Secured **All India Rank 1540** in JEE Advanced (top 0.9% among 160,000+ candidates).
- Recipient of the Merit-cum-Means Scholarship at IIT Delhi across all eight academic semesters.